Section 3–5 presents concepts of work, energy, and power. This section also includes a presentation of an energy method for obtaining equations of motion of conservative systems. Finally, Section 3–6 discusses mechanical motion, energy, and power transformers.

3–2 MECHANICAL ELEMENTS

Any mechanical system consists of mechanical elements. There are three types of basic elements in mechanical systems: inertia elements, spring elements, and damper elements.

Inertia elements. By *inertia elements* we mean masses and moments of inertia.

Inertia may be defined as the change in force (torque) required to make a unit change in acceleration (angular acceleration). That is,

$$\text{inertia (mass)} = \frac{\text{change in force}}{\text{change in acceleration}} \quad \frac{\text{N}}{\text{m/s}^2} \text{ or kg}$$

$$\text{inertia (moment of inertia)} = \frac{\text{change in torque}}{\text{change in angular acceleration}} \quad \frac{\text{N-m}}{\text{rad/s}^2} \text{ or kg-m}^2$$

Spring elements. A linear *spring* is a mechanical element that can be deformed by an external force such that the deformation is directly proportional to the force or torque applied to it.

Figure 3–1 is a schematic diagram of a spring. Here we consider translational motion only. The spring has been deflected from its initial position by a force applied to each end. Positions x_1 and x_2 of the ends of the spring are measured relative to the same frame of reference. The forces at both ends of the spring are on the same line and are equal in magnitude. Then force F and net displacement x of the ends of the spring are related by

$$F = kx = k(x_1 - x_2) \tag{3–1}$$

where k is a proportionality constant called a *spring constant.* The dimension of spring constant k is force/displacement.

For rotational motion, the torque applied to the ends of a torsional or rotational spring and the net change in the angular displacements of the ends are related by

$$T = k\theta = k(\theta_1 - \theta_2) \tag{3–2}$$

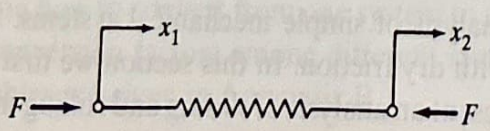


Figure 3–1 Spring.

where

T = torque applied to the ends of torsional spring
θ_1 = angular displacement of one end
θ_2 = angular displacement of other end
$\theta = \theta_1 - \theta_2$ = net angular displacement of ends
k = torsional spring constant

The dimension of torsional spring constant k is torque/angular displacement, where angular displacement is measured in radians.

When a linear spring is stretched, a point is reached in which the force per unit displacement begins to change and the spring becomes a nonlinear spring. If stretched farther, a point is reached at which the material will either break or yield. For practical springs, therefore, the assumption of linearity may be good only for relatively small net displacements. Figure 3–2 shows the force–displacement characteristic curves for linear and nonlinear springs.

For practical springs, spring constant k (or an equivalent spring constant for a nonlinear spring) may be defined as

$$\text{spring constant } k \text{ (for translational spring)} = \frac{\text{change in force}}{\text{change in displacement of spring}} \quad \frac{\text{N}}{\text{m}}$$

$$\text{spring constant } k \text{ (for torsional spring)} = \frac{\text{change in torque}}{\text{change in angular displacement of spring}} \quad \frac{\text{N-m}}{\text{rad}}$$

Spring constants indicate stiffness; a large value of k corresponds to a hard spring and a small value of k to a soft spring. The reciprocal of spring constant k is called *compliance* or *mechanical capacitance* C. Thus $C = 1/k$. Compliance or mechanical capacitance indicates the softness of a spring.

Practical spring versus ideal spring. All practical springs have inertia and damping. In our analysis in this book, however, we assume that the effect of the mass in a spring is negligibly small; that is, the inertia force due to acceleration of the spring is negligibly small compared with the spring force. Also, we assume that the damping effect of the spring is negligibly small.

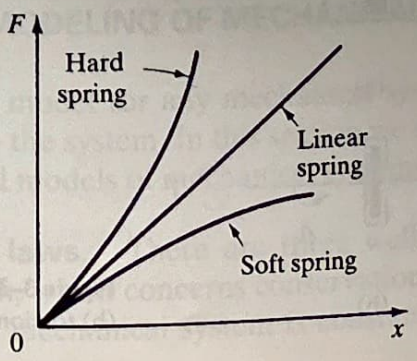


Figure 3–2 Force–displacement characteristic curves for linear and nonlinear springs.

An ideal spring, in comparison to a practical spring, will have neither mass nor damping (internal friction) and will obey the linear force–displacement law as given by Equation (3–1) or the linear torque–angular displacement law as given by Equation (3–2).

Damper elements. A *damper* is a mechanical element that dissipates energy in the form of heat instead of storing it. Figure 3–3(a) shows a schematic diagram of a translational damper or dashpot. It consists of a piston and an oil-filled cylinder. Any relative motion between the piston rod and the cylinder is resisted by oil because oil must flow around the piston (or through orifices provided in the piston) from one side to the other. Essentially, the damper absorbs energy, and the absorbed energy is dissipated as heat that flows away to the surroundings.

In Figure 3–3(a), velocities $\dot{x}_1$ and $\dot{x}_2$ are taken relative to the same frame of reference. The forces at the ends of the translational damper are on the same line and are equal in magnitude. In the damper, the force F acting on it is proportional to the velocity difference $\dot{x}$ of both ends, or

$$F = b\dot{x} = b(\dot{x}_1 - \dot{x}_2) \tag{3–3}$$

where proportionality constant b relating external force F and velocity difference $\dot{x}$ is called a *viscous friction coefficient* or *viscous friction constant.* The dimension of the viscous friction coefficient is force/velocity. Note that the initial positions of both ends of the damper do not appear in the equation.

For the torsional or rotational damper shown in Figure 3–3(b), the torque applied to the ends of the damper is proportional to the angular velocity difference of both ends, or

$$T = b\dot{\theta} = b(\dot{\theta}_1 - \dot{\theta}_2) \tag{3–4}$$

where

T = torque applied to the ends of torsional damper
$\dot{\theta}_1$ = angular velocity of one end
$\dot{\theta}_2$ = angular velocity of other end
$\dot{\theta} = \dot{\theta}_1 - \dot{\theta}_2$ = relative angular velocity of ends
b = torsional viscous friction coefficient (torsional viscous friction constant)

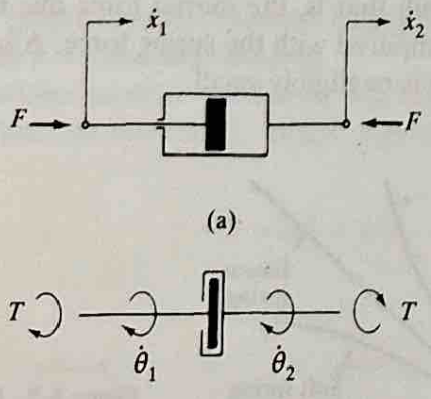


Figure 3–3 (a) Translational damper; (b) torsional (or rotational) damper.

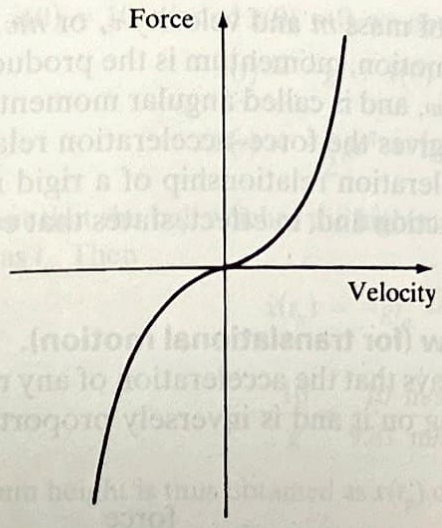


Figure 3–4 Characteristic curve for square-law friction.

The dimension of the torsional viscous friction coefficient b is torque/angular velocity.

Note that a damper is an element that provides resistance in mechanical motion and, as such, its effect on the dynamic behavior of a mechanical system is similar to that of an electrical resistor on the dynamic behavior of an electrical system. Consequently, a damper is often referred to as a *mechanical resistance element* and the viscous friction coefficient as the *mechanical resistance.*

Practical damper versus ideal damper. All practical dampers produce inertia and spring effects. In this book, however, we assume that these effects are negligible.

An ideal damper is massless and springless, dissipates all energy, and obeys the linear force–velocity law or linear torque–angular velocity law as given by Equation (3–3) or Equation (3–4), respectively.

Nonlinear friction. Friction that obeys linear law is called *linear friction,* whereas friction that does not is described as *nonlinear.* Examples of nonlinear friction include static friction, sliding friction, and square-law friction.The topics of static friction and sliding friction are discussed in Section 3–4.

Square-law friction occurs when a solid body moves in a fluid medium. Figure 3–4 shows a characteristic curve for square-law friction.

3–3 MATHEMATICAL MODELING OF MECHANICAL SYSTEMS

A mathematical model for any mechanical system can be developed by applying Newton's laws to the system. In this section we shall deal with the problem of deriving mathematical models of mechanical systems.

Newton's laws. There are three well-known laws called *Newton's laws.* Newton's first law, which concerns conservation of momentum, states that the total momentum of a mechanical system is constant in the absence of external forces.

Momentum is the product of mass m and velocity v, or mv, for translational or linear motion. For rotational motion, momentum is the product of moment of inertia J and angular velocity ω, or $J\omega$, and is called angular momentum.

Newton's second law gives the force–acceleration relationship of a rigid body or the torque–angular acceleration relationship of a rigid rotating body. The third law concerns action and reaction and, in effect, states that every action is always opposed by an equal reaction.

Newton's second law (for translational motion). For a translational motion, Newton's second law says that the acceleration of any rigid body is directly proportional to the force acting on it and is inversely proportional to the mass of the body. That is,

$$\text{acceleration} = \frac{\text{force}}{\text{mass}}$$

or

$$(\text{mass})(\text{acceleration}) = \text{force}$$

Suppose that forces are acting on a mass m. If ΣF is the sum of all forces acting in a given direction, then

$$ma = \sum F \tag{3–5}$$

where a is the resulting absolute acceleration in that direction. The line of action of the force acting on a mass must pass through the center of mass. Otherwise, rotational motion will also be involved. Rotational motion is not defined by Equation (3–5) but is given by Equation (3–6), which follows.

Newton's second law (for rotational motion). For a rigid body in pure rotation about a fixed axis, Newton's second law states that

$$(\text{moment of inertia})(\text{angular acceleration}) = \text{torque}$$

or

$$J\alpha = \sum T \tag{3–6}$$

where ΣT is the sum of all torques acting about a given axis, J is the moment of inertia of a body about that axis, and α is the angular acceleration.

Example 3–1

A ball of mass m is thrown vertically upward with an initial velocity of 10 m/s. The vertical displacement x is measured upward from the starting point. What is the maximum height that the ball will reach? Assume that air friction is negligible.

The only force acting on the ball is the gravitational force $-mg$. Applying Newton's second law, we have

$$m\ddot{x} = -mg$$

Noting that $\dot{x}(0) = 10$ m/s and $x(0) = 0$, we obtain

$$\dot{x}(t) = -gt + \dot{x}(0) = -gt + 10$$

$$x(t) = -\frac{1}{2}gt^2 + 10t$$

At the instant that the ball reaches the highest point the velocity is zero. Let us define this instant as t_p. Then

$$\dot{x}(t_p) = -gt_p + 10 = 0$$

or

$$t_p = \frac{10}{g} = \frac{10 \text{ m/s}}{9.81 \text{ m/s}^2} = 1.02 \text{ s}$$

The maximum height is thus obtained as $x(t_p)$ or

$$x(t_p) = -\frac{1}{2}gt_p^2 + 10t_p$$

$$= -\frac{1}{2} \times 9.81 \times 1.02^2 + 10 \times 1.02 = 5.10$$

And so the maximum height that the ball will reach is 5.10 m.

Torque or moment of force. *Torque,* or moment of force, is defined as any cause that tends to produce a change in the rotational motion of a body on which it acts. Torque is the product of a force and the perpendicular distance from a point of rotation to the action line of the force. The units of torque are units of force times length, such as N-m, dyn-cm, kg_f-m, and lb_f-ft.

Rigid body. When any real body is accelerated, internal elastic deflections are always present. If these internal deflections are negligibly small relative to the gross motion of the entire body, the body is called a *rigid body.* Thus a rigid body does not deform. In other words, for a pure translational motion, every point in a rigid body has identical motion.

Moments of inertia. The *moment of inertia J* of a rigid body about an axis is defined by

$$J = \int r^2 \, dm$$

where dm is an element of mass, r is distance from the axis to dm, and integration is performed over the body. In considering moments of inertia, we assume that the rotating body is perfectly rigid. Physically, the moment of inertia of a body is a measure of its resistance to angular acceleration.

Consider a rectangular (xyz) coordinate system in a body. The moment of inertia of a body about the x axis is

$$J = \int (y^2 + z^2) \, dm$$

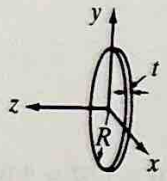
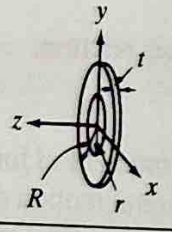
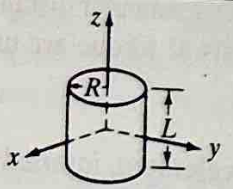
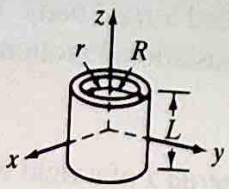
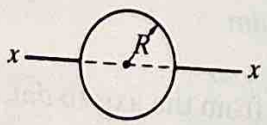
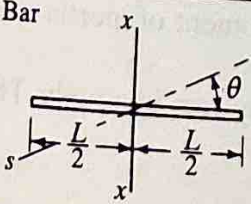
Similarly, those about the y and z axes are

$$J_y = \int (z^2 + x^2)\, dm$$

$$J_z = \int (x^2 + y^2)\, dm$$

Table 3–3 gives the moments of inertia of rigid bodies in common shapes.

TABLE 3–3 Moments of Inertia

	$t \ll R$ M = mass $J_x = J_y = M\dfrac{R^2}{4}$, $J_z = J_x + J_y = M\dfrac{R^2}{2}$
	$t \ll R$ M = mass $J_x = J_y = M\dfrac{R^2 + r^2}{4}$, $J_z = M\dfrac{R^2 + r^2}{2}$
	M = mass $J_x = J_y = M\dfrac{3R^2 + L^2}{12}$, $J_z = M\dfrac{R^2}{2}$
	M = mass $J_x = J_y = M\dfrac{R^2 + r^2 + \frac{1}{3}L^2}{4}$, $J_z = M\dfrac{R^2 + r^2}{2}$
Sphere 	M = mass ρ = density $J_{xx} = \dfrac{2}{5}MR^2$ $\left(M = \dfrac{4}{3}\pi R^3 \rho\right)$
Bar 	M = mass $J_{xx} = M\dfrac{L^2}{12}$, $J_{ss} = M\dfrac{(L\sin\theta)^2}{12}$

Example 3–2

Figure 3–5 shows a homogeneous cylinder of radius R and length L. The moment of inertia of this cylinder about axis AA' can be obtained as follows. Consider a ring-shaped mass element of infinitesimal width dr at radius r. The mass of this ring-shaped element is $2\pi rL\rho\, dr$, where ρ is the density of this cylinder. Thus

$$dm = 2\pi rL\rho dr$$

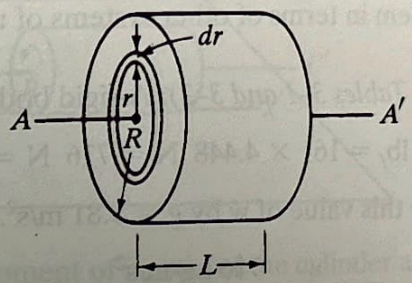


Figure 3–5 Homogeneous cylinder.

Consequently,

$$J = \int_0^R r^2 2\pi rL\rho\, dr = 2\pi L\rho \int_0^R r^3\, dr = \frac{\pi L\rho R^4}{2}$$

Since the entire mass m of the cylinder body is $m = \pi R^2 L\rho$, we obtain

$$J = \frac{1}{2}mR^2$$

Radius of gyration. The *radius of gyration* of a rigid body with reference to an axis is a length k, which, when squared and multiplied by the mass m of a rigid body, is equal to the moment of inertia J of the body with reference to the same inertia axis, or

$$mk^2 = J$$

Therefore,

$$k = \sqrt{\frac{J}{m}}$$

Physically, if it were possible to concentrate all the mass of the rigid body at a point of some distance k from the axis in question, so that mk^2 is equal to the moment of inertia J of the body with respect to that axis, then the distance k is the radius of gyration.

Example 3–3

Suppose that a rigid body weighing 161 lb_f has a moment of inertia about a particular axis of 125 slug-ft^2. What is the radius of gyration about this axis?

The mass of the rigid body is

$$m = \frac{w}{g} = \frac{161}{32.2} = 5 \text{ slugs}$$

The radius of gyration k is

$$k = \sqrt{\frac{J}{m}} = \sqrt{\frac{125}{5}} = 5 \text{ ft}$$

In this example we used units of the British engineering system. As an exercise, let us interpret this problem in terms of other systems of units.

1. SI units (refer to Tables 3–1 and 3–2): A rigid body weighs

$$w = 161 \text{ lb}_f = 161 \times 4.448 \text{ N} = 716 \text{ N} = 716 \text{ kg-m/s}^2$$

To get mass m, we divide this value of w by $g = 9.81 \text{ m/s}^2$.

$$m = \frac{w}{g} = \frac{716 \text{ kg-m/s}^2}{9.81 \text{ m/s}^2} = 73.0 \text{ kg}$$

The moment of inertia J becomes

$$J = 125 \text{ slug-ft}^2 = 125 \times 1.356 \text{ kg-m}^2 = 169.5 \text{ kg-m}^2$$

Hence the radius of gyration k is

$$k = \sqrt{\frac{J}{m}} = \sqrt{\frac{169.5}{73.0}} = 1.524 \text{ m}$$

2. Metric engineering (gravitational) units (refer to Tables 3–1 and 3–2): A rigid body weighs

$$w = 161 \text{ lb}_f = 161 \times 0.4536 \text{ kg}_f = 73.0 \text{ kg}_f$$

Mass m is then obtained as

$$m = \frac{w}{g} = \frac{73.0 \text{ kg}_f}{9.81 \text{ m/s}^2} = 7.44 \text{ kg}_f\text{-s}^2/\text{m}$$

Also, the moment of inertia J is

$$J = 125 \text{ slug-ft}^2 = 125 \times 0.1383 \text{ kg}_f\text{-s}^2\text{-m} = 17.28 \text{ kg}_f\text{-s}^2\text{-m}$$

Consequently,

$$k = \sqrt{\frac{J}{m}} = \sqrt{\frac{17.28}{7.44}} = 1.524 \text{ m}$$

Moment of inertia about an axis other than the geometrical axis. Sometimes it is necessary to calculate the moment of inertia of a homogeneous rigid body about an axis other than its geometrical axis. If the axes are parallel, the calculation can be done easily. The moment of inertia about an axis that is a distance x from the geometrical axis passing through the center of gravity is the sum of the mo-

ment of inertia about the geometrical axis and the moment of inertia about the new axis when the mass of the body is considered concentrated at the center of gravity.

Example 3–4

Consider the system shown in Figure 3–6, where a homogeneous cylinder of mass m and radius R rolls on a flat surface. Find the moment of inertia J_x of the cylinder about its line of contact (axis xx') with the surface.

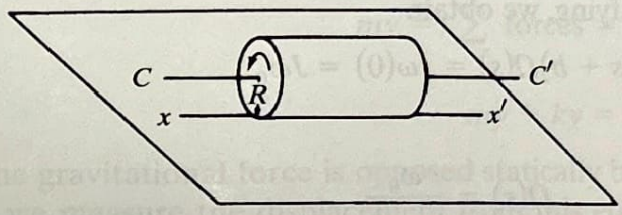


Figure 3–6 Homogeneous cylinder rolling on a flat surface.

The moment of inertia of the cylinder about axis CC' is

$$J_C = \frac{1}{2}mR^2$$

The moment of inertia of the cylinder about axis xx' when mass m is considered concentrated at the center of gravity is mR^2. Thus the moment of inertia J_x of the cylinder about axis xx' is

$$J_x = J_C + mR^2 = \frac{1}{2}mR^2 + mR^2 = \frac{3}{2}mR^2$$

Forced response and natural response. The behavior determined by a forcing function is called a *forced response* and that due to initial conditions (initial energy storages) is called a *natural response.* The period between the initiation of a natural response and the ending is referred to as the *transient period.* After the natural response has become negligibly small, conditions are said to have reached a steady state.

Rotational systems. A schematic diagram of a rotor mounted in bearings is shown in Figure 3–7. The moment of inertia of the rotor about the axis of rotation is J. Let us assume that at $t = 0$ the rotor is rotating at the angular velocity of $\omega(0) = \omega_0$. We also assume that the friction in the bearings is viscous friction and that no external torque is applied to the rotor. Then the only torque acting on the rotor is the friction torque $b\omega$ in the bearings.

Applying Newton's second law, Equation (3–6), we obtain the equation of motion:

$$J\dot{\omega} = -b\omega, \qquad \omega(0) = \omega_0$$

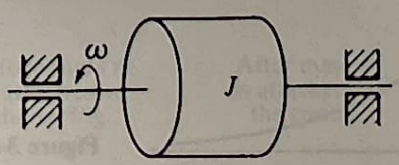


Figure 3–7 Rotor mounted in bearings.

or

$$J\dot{\omega} + b\omega = 0 \tag{3-7}$$

Equation (3–7) is the equation of motion, as well as a mathematical model, of the system. A system governed by a first-order differential equation, such as this one, is called a *first-order system.*

To find $\omega(t)$, we take the Laplace transform of Equation (3–7).

$$J[s\Omega(s) - \omega(0)] + b\Omega(s) = 0$$

where $\Omega(s) = \mathcal{L}[\omega(t)]$. Simplifying, we obtain

$$(Js + b)\Omega(s) = J\omega(0) = J\omega_0$$

Hence

$$\Omega(s) = \frac{\omega_0}{s + \dfrac{b}{J}} \tag{3-8}$$

The denominator $s + (b/J)$ is called the characteristic polynomial and

$$s + \frac{b}{J} = 0$$

is called the characteristic equation.

The inverse Laplace transform of $\Omega(s)$, the solution of the differential equation given by Equation (3–7), is

$$\omega(t) = \omega_0 e^{-(b/J)t}$$

The angular velocity decreases exponentially, as shown in Figure 3–8.

Since the exponential factor $e^{-(b/J)t}$ approaches zero as t increases without limit, mathematically the response lasts forever. When dealing with such an exponentially decaying response, it is convenient to depict the response in terms of a time constant. A *time constant* is that value of time that makes the exponent equal to -1. For this system, time constant T is equal to J/b, or $T = J/b$. When $t = T$, the value of the exponential factor is

$$e^{-T/T} = e^{-1} = 0.368$$

In other words, when time t in seconds is equal to the time constant, the exponential factor is reduced to approximately 36.8% of its initial value, as shown in Figure 3–8.

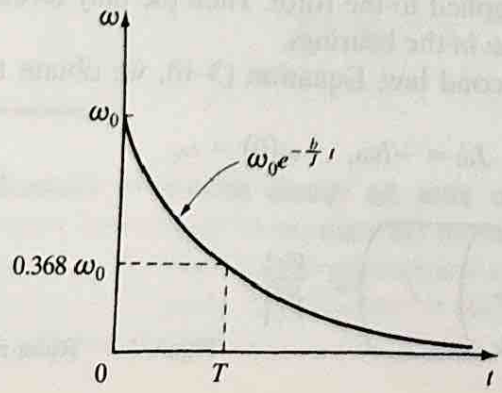


Figure 3–8 Angular velocity curve for the rotor system shown in Figure 3–7.

Spring–mass system. Figure 3–9 depicts a system consisting of a mass and a spring. Here the mass is suspended by the spring. For the vertical motion, two forces are acting on the mass, spring force ky and gravitational force mg. (In the diagram the positive direction of displacement y is defined downward.) If the mass is pulled downward by an external force and then released, the spring force acts upward and tends to pull the mass upward. The gravitational force pulls the mass downward. So, by applying Newton's second law to this system, we obtain the following equation of motion:

$$m\ddot{y} = \sum \text{forces} = -ky + mg$$

or

$$m\ddot{y} + ky = mg \tag{3–9}$$

The gravitational force is opposed statically by the equilibrium spring deflection δ. If we measure the displacement from this equilibrium position, then the term mg can be dropped from the equation of motion. By substituting $y = x + \delta$ into Equation (3–9) and noting that δ = constant, we have

$$m\ddot{x} + k(x + \delta) = mg$$

Since $k\delta = mg$, this last equation simplifies to

$$m\ddot{x} + kx = 0 \tag{3–10}$$

which is a mathematical model for the system. Notice that the dynamics of this system is governed by a second-order differential equation. Such a system is called a second-order system.

Unless otherwise stated, when writing equations of motion for systems involving the gravitational force in this book, we measure the displacement of the mass from the equilibrium position in order to eliminate the term mg and simplify the mathematical model.

Free vibration. For the spring–mass system of Figure 3–9, suppose that the mass is pulled downward and then released with arbitrary initial conditions $x(0)$ and

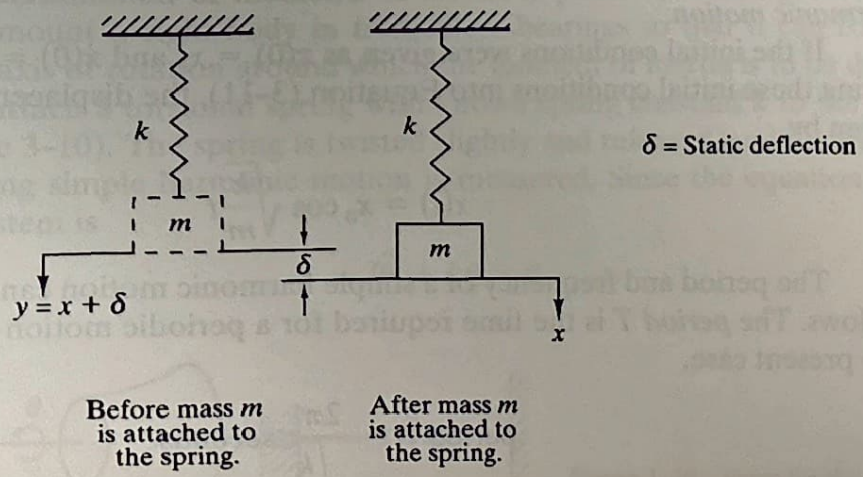


Figure 3–9 Spring–mass system.

or

$$J\dot{\omega} + b\omega = 0 \tag{3–7}$$

Equation (3–7) is the equation of motion, as well as a mathematical model, of the system. A system governed by a first-order differential equation, such as this one, is called a *first-order system.*

To find $\omega(t)$, we take the Laplace transform of Equation (3–7).

$$J[s\Omega(s) - \omega(0)] + b\Omega(s) = 0$$

where $\Omega(s) = \mathcal{L}[\omega(t)]$. Simplifying, we obtain

$$(Js + b)\Omega(s) = J\omega(0) = J\omega_0$$

Hence

$$\Omega(s) = \frac{\omega_0}{s + \dfrac{b}{J}} \tag{3–8}$$

The denominator $s + (b/J)$ is called the characteristic polynomial and

$$s + \frac{b}{J} = 0$$

is called the characteristic equation.

The inverse Laplace transform of $\Omega(s)$, the solution of the differential equation given by Equation (3–7), is

$$\omega(t) = \omega_0 e^{-(b/J)t}$$

The angular velocity decreases exponentially, as shown in Figure 3–8.

Since the exponential factor $e^{-(b/J)t}$ approaches zero as t increases without limit, mathematically the response lasts forever. When dealing with such an exponentially decaying response, it is convenient to depict the response in terms of a time constant. A *time constant* is that value of time that makes the exponent equal to -1. For this system, time constant T is equal to J/b, or $T = J/b$. When $t = T$, the value of the exponential factor is

$$e^{-T/T} = e^{-1} = 0.368$$

In other words, when time t in seconds is equal to the time constant, the exponential factor is reduced to approximately 36.8% of its initial value, as shown in Figure 3–8.

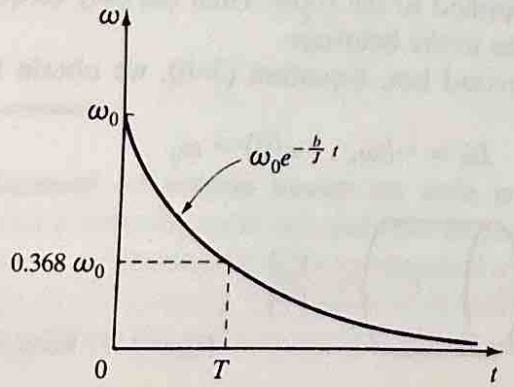


Figure 3–8 Angular velocity curve for the rotor system shown in Figure 3–7.

Spring–mass system. Figure 3–9 depicts a system consisting of a mass and a spring. Here the mass is suspended by the spring. For the vertical motion, two forces are acting on the mass, spring force ky and gravitational force mg. (In the diagram the positive direction of displacement y is defined downward.) If the mass is pulled downward by an external force and then released, the spring force acts upward and tends to pull the mass upward. The gravitational force pulls the mass downward. So, by applying Newton's second law to this system, we obtain the following equation of motion:

$$m\ddot{y} = \sum \text{forces} = -ky + mg$$

or

$$m\ddot{y} + ky = mg \tag{3–9}$$

The gravitational force is opposed statically by the equilibrium spring deflection δ. If we measure the displacement from this equilibrium position, then the term mg can be dropped from the equation of motion. By substituting $y = x + \delta$ into Equation (3–9) and noting that δ = constant, we have

$$m\ddot{x} + k(x + \delta) = mg$$

Since $k\delta = mg$, this last equation simplifies to

$$m\ddot{x} + kx = 0 \tag{3–10}$$

which is a mathematical model for the system. Notice that the dynamics of this system is governed by a second-order differential equation. Such a system is called a second-order system.

Unless otherwise stated, when writing equations of motion for systems involving the gravitational force in this book, we measure the displacement of the mass from the equilibrium position in order to eliminate the term mg and simplify the mathematical model.

Free vibration. For the spring–mass system of Figure 3–9, suppose that the mass is pulled downward and then released with arbitrary initial conditions $x(0)$ and

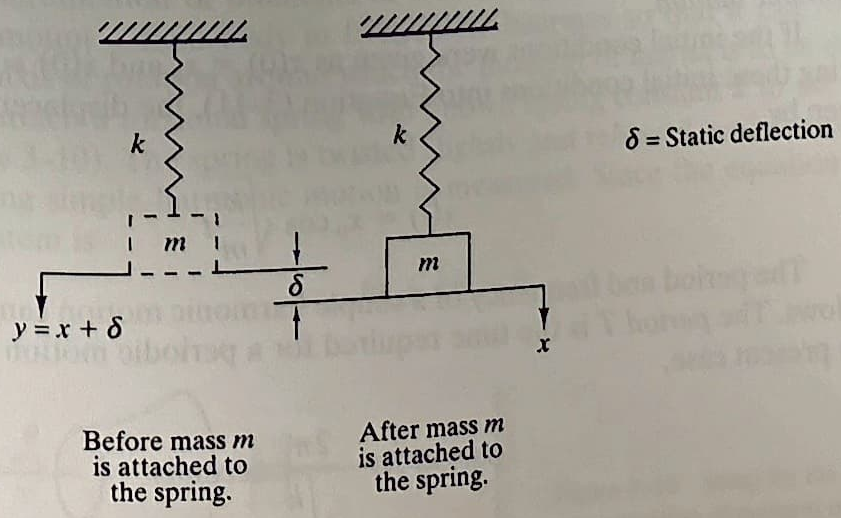


Figure 3–9 Spring–mass system.

$\dot{x}(0)$. In this case, the mass will oscillate and the motion will be periodic. The periodic motion, observed as the system is displaced from its static equilibrium position, is called *free vibration.* It is a natural response due to the initial condition.

To find the mathematical form of the periodic motion, let us solve Equation (3–10). By taking the Laplace transforms of both sides of Equation (3–10), we obtain

$$m[s^2X(s) - sx(0) - \dot{x}(0)] + kX(s) = 0$$

or

$$(ms^2 + k)X(s) = m\dot{x}(0) + msx(0)$$

Hence

$$X(s) = \frac{\dot{x}(0)}{s^2 + \dfrac{k}{m}} + \frac{sx(0)}{s^2 + \dfrac{k}{m}}$$

This last equation may be rewritten so that the inverse Laplace transform of each term can be easily identified.

$$X(s) = \sqrt{\frac{m}{k}}\dot{x}(0)\frac{\sqrt{k/m}}{s^2 + (\sqrt{k/m})^2} + x(0)\frac{s}{s^2 + (\sqrt{k/m})^2}$$

Noting that

$$\mathcal{L}[\sin \sqrt{k/m}\, t] = \frac{\sqrt{k/m}}{s^2 + (\sqrt{k/m})^2}$$

$$\mathcal{L}[\cos \sqrt{k/m}\, t] = \frac{s}{s^2 + (\sqrt{k/m})^2}$$

the inverse Laplace transform of $X(s)$ is obtained as

$$x(t) = \sqrt{\frac{m}{k}}\dot{x}(0)\sin\sqrt{\frac{k}{m}}\,t + x(0)\cos\sqrt{\frac{k}{m}}\,t \qquad (3\text{–}11)$$

The periodic motion such as that described by Equation (3–11) is called a *simple harmonic motion.*

If the initial conditions were given as $x(0) = x_0$ and $\dot{x}(0) = 0$, then, by substituting these initial conditions into Equation (3–11), the displacement $x(t)$ would be given by

$$x(t) = x_0 \cos\sqrt{\frac{k}{m}}\,t$$

The period and frequency of a simple harmonic motion can now be defined as follows. The *period* T is the time required for a periodic motion to repeat itself. In the present case,

$$\text{period } T = \frac{2\pi}{\sqrt{\dfrac{k}{m}}} \text{ seconds}$$

The *frequency* f of a periodic motion is the number of cycles per second, and the standard unit of frequency is the hertz (Hz); that is, one hertz is one cycle per second. In the present harmonic motion,

$$\text{frequency } f = \frac{1}{T} = \frac{\sqrt{\frac{k}{m}}}{2\pi} \text{ hertz}$$

The natural frequency or undamped natural frequency is the frequency in the free vibration of a system having no damping. If the natural frequency is measured in hertz (Hz) or cycles per second (cps), it is denoted by f_n. If it is measured in radians per second (rad/s), it is denoted by ω_n. In the present system

$$\omega_n = 2\pi f_n = \sqrt{\frac{k}{m}} \text{ rad/s}$$

It is important to remember that, when Equation (3–10) is written in a form such that the coefficient of the $\ddot{x}$ term is unity,

$$\ddot{x} + \frac{k}{m}x = 0$$

the square root of the coefficient of the x term is the natural frequency ω_n. This means that a mathematical model for the system shown in Figure 3–9 can be put in the form

$$\ddot{x} + \omega_n^2 x = 0$$

where $\omega_n = \sqrt{k/m}$.

Experimental determination of moment of inertia. It is possible to calculate moments of inertia for homogeneous bodies having geometrically simple shapes. However, for rigid bodies with complicated shapes or those consisting of materials of various densities, such calculation may be difficult or even impossible; moreover, calculated values may not be accurate. In these instances, experimental determination of moments of inertia is preferable. The process is as follows. We mount a rigid body in frictionless bearings so that it can rotate freely about the axis of rotation around which the moment of inertia is to be determined. Next, we attach a torsional spring with known spring constant k to the rigid body (see Figure 3–10). The spring is twisted slightly and released, and the frequency of the resulting simple harmonic motion is measured. Since the equation of motion for this system is

$$J\ddot{\theta} + k\theta = 0$$

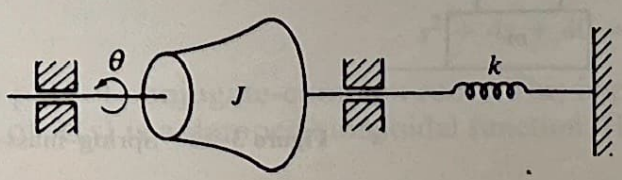


Figure 3–10 Setup for the experimental determination of moment of inertia.

or

$$\ddot{\theta} + \frac{k}{J}\theta = 0$$

the natural frequency ω_n is

$$\omega_n = \sqrt{\frac{k}{J}}$$

and the period of vibration is

$$T = \frac{2\pi}{\omega_n} = \frac{2\pi}{\sqrt{\frac{k}{J}}}$$

The moment of inertia J is then determined as

$$J = \frac{kT^2}{4\pi^2}$$

Similarly, in the spring–mass system of Figure 3–9, if the spring constant k is known and the period T of the free vibration is measured, then mass m can be calculated from

$$m = \frac{kT^2}{4\pi^2}$$

Spring–mass–damper system. Most physical systems involve some type of damping—viscous damping, magnetic damping, and so on. Such damping not only slows the motion, but also causes the motion to stop eventually. In the following discussion we shall consider a simple mechanical system involving viscous damping. Note that a typical viscous damping element is a damper or dashpot.

Figure 3–11 is a schematic diagram of a spring–mass–damper system. Suppose that the mass is pulled downward and then released. If the damping is light, vibratory motion will occur. (The system is said to be underdamped.) If the damping is heavy, vibratory motion will not occur. (The system is said to be overdamped.) A critically damped system is one in which the degree of damping is such that the resultant motion is on the borderline between the underdamped and overdamped

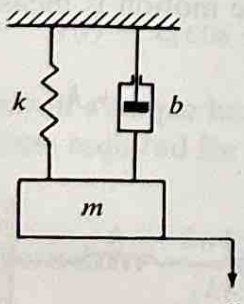


Figure 3–11 Spring–mass–damper system.

cases. Regardless of the system being underdamped, overdamped, or critically damped, because of the presence of damper the free vibration or free motion will diminish with time. This free vibration is called *transient.*

In the system shown in Figure 3–11, for the vertical motion, three forces are acting on the mass: spring force, damping force, and gravitational force. As noted earlier, if we measure the displacement of the mass from a static equilibrium position (so that the gravitational force is balanced by the equilibrium spring deflection), the gravitational force will not enter the equation of motion. So by measuring the displacement x from the static equilibrium position, we obtain the equation of motion

$$m\ddot{x} = \sum \text{forces} = -kx - b\dot{x}$$

or

$$m\ddot{x} + b\dot{x} + kx = 0 \tag{3–12}$$

Equation (3–12), which describes the motion of the system, is a mathematical model of the system.

Only the underdamped case is considered in our present analysis. (A more complete analysis of this system for the underdamped, overdamped, and critically damped cases is given in Chapter 6.)

Let us solve Equation (3–12) for a particular case. Suppose that $m = 0.1$ slug, $b = 0.4$ lb$_f$-s/ft, and $k = 4$ lb$_f$/ft. Then Equation (3–12) becomes

$$0.1\ddot{x} + 0.4\dot{x} + 4x = 0$$

or

$$\ddot{x} + 4\dot{x} + 40x = 0 \tag{3–13}$$

Let us obtain the motion $x(t)$ when the mass is pulled downward at $t = 0$, such that $x(0) = x_0$, and released with zero velocity, $\dot{x}(0) = 0$. Taking the Laplace transform of Equation (3–13), we obtain

$$[s^2X(s) - sx(0) - \dot{x}(0)] + 4[sX(s) - x(0)] + 40X(s) = 0$$

Simplifying this last equation, noting that $x(0) = x_0$ and $\dot{x}(0) = 0$, we get

$$(s^2 + 4s + 40)X(s) = sx_0 + 4x_0$$

or

$$X(s) = \frac{(s + 4)x_0}{s^2 + 4s + 40}$$

The characteristic equation for the system

$$s^2 + 4s + 40 = 0$$

has a pair of conjugate-complex roots. This implies that the inverse Laplace transform of $X(s)$ is a damped sinusoidal function. Hence we may rewrite $X(s)$ as a sum

of the Laplace transforms of a damped sine function and a damped cosine function as follows:

$$X(s) = \frac{1}{3}x_0\frac{6}{(s+2)^2+6^2} + x_0\frac{s+2}{(s+2)^2+6^2}$$

Noting that

$$\frac{6}{(s+2)^2+6^2} = \mathcal{L}[e^{-2t}\sin 6t], \qquad \frac{s+2}{(s+2)^2+6^2} = \mathcal{L}[e^{-2t}\cos 6t]$$

the inverse Laplace transform of $X(s)$ can be obtained as

$$\begin{aligned} x(t) &= \frac{1}{3}x_0e^{-2t}\sin 6t + x_0e^{-2t}\cos 6t \\ &= e^{-2t}\left(\frac{1}{3}\sin 6t + \cos 6t\right)x_0 \end{aligned} \tag{3–14}$$

Equation (3–14) depicts the free vibration of the spring–mass–damper system with the given numerical values. The free vibration here is a damped sinusoidal vibration, as shown in Figure 3–12.

Comments. The numerical values in the preceding problem were stated in BES units. Let us convert these values into units of other systems.

1. SI units (refer to Tables 3–1 and 3–2):

$$m = 0.1 \text{ slug} = 1.459 \text{ kg}$$
$$b = 0.4 \text{ lb}_f\text{-s/ft} = 0.4 \times 4.448/0.3048 \text{ N-s/m} = 5.837 \text{ N-s/m}$$
$$k = 4 \text{ lb}_f\text{/ft} = 4 \times 4.448/0.3048 \text{ N/m} = 58.37 \text{ N/m}$$

Hence Equation (3–12) becomes

$$1.459\ddot{x} + 5.837\dot{x} + 58.37x = 0$$

or

$$\ddot{x} + 4\dot{x} + 40x = 0$$

which is the same as Equation (3–13).

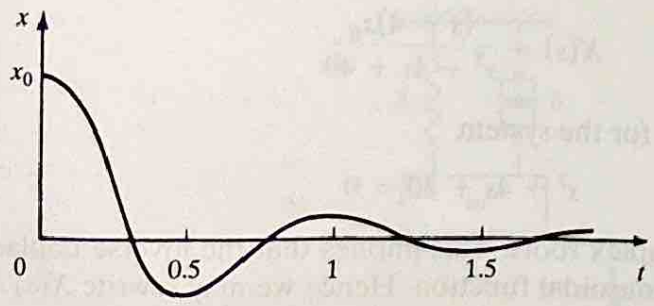


Figure 3–12 Free vibration of the spring–mass–damper system described by $\ddot{x} + 4\dot{x} + 40x = 0$ with initial conditions $x(0) = x_0$ and $\dot{x}(0) = 0$.

2. Metric engineering (gravitational) units (refer to Tables 3–1 and 3–2):

$$m = 0.1 \text{ slug} = 0.1488 \text{ kg}_f\text{-s}^2/\text{m}$$
$$b = 0.4 \text{ lb}_f\text{-s/ft} = 0.4 \times 0.4536/0.3048 \text{ kg}_f\text{-s/m} = 0.5953 \text{ kg}_f\text{-s/m}$$
$$k = 4 \text{ lb}_f/\text{ft} = 4 \times 0.4536/0.3048 \text{ kg}_f/\text{m} = 5.953 \text{ kg}_f/\text{m}$$

Therefore, Equation (3–12) becomes

$$0.1488\ddot{x} + 0.5953\dot{x} + 5.953x = 0$$

or

$$\ddot{x} + 4\dot{x} + 40x = 0$$

which is the same as Equation (3–13).

Note that as long as we use consistent units the differential equation (mathematical model) of the system remains the same.

3–4 MECHANICAL SYSTEMS WITH DRY FRICTION

Sliding, rolling, and rubbing of various parts constitute some of the friction forces involved in mechanical systems. In most cases, actual friction forces are a combination of viscous friction, dry friction, and various other types.

In this section we are concerned with *dry friction,* the friction force observed when a body with an unlubricated surface slides over another unlubricated surface. We begin with static friction, sliding friction, and rolling friction. Afterward we derive mathematical models of mechanical systems with dry friction, followed by response analysis of such systems.

Static friction and sliding friction. Whenever the surface of one body slides over that of another, each body exerts a frictional force on the other, parallel to the surfaces. The force on each body is opposite to the direction of its motion relative to the other.

Suppose that a body is placed on a rough surface and that a pulling force is exerted on it [see Figure 3–13(a)]. If the body is pulled with an increasing force, it will not move at first. But as the magnitude of the force increases and reaches a sufficient value to overcome the friction between two contacting surfaces, the body will begin to move. When two contacting surfaces are at rest relative to each other, the static friction force reaches a maximum when sliding between the two surfaces is imminent. Immediately after the motion begins, the magnitude of the friction force decreases slightly. The friction force acting on the body when it is moving with uniform motion is called *sliding* or *kinetic friction.* Sometimes it is also referred to as *coulomb friction.* A characteristic curve for static and sliding friction appears in Figure 3–13(b).

For the system shown in Figure 3–13(a), the forces acting on the body, other than the pulling and friction forces, are the gravitational force and the so-called normal force, which is created by the surface on which the body is resting or sliding and which is pushing up on the body. The latter force acts *normal* to the surface, hence its name.

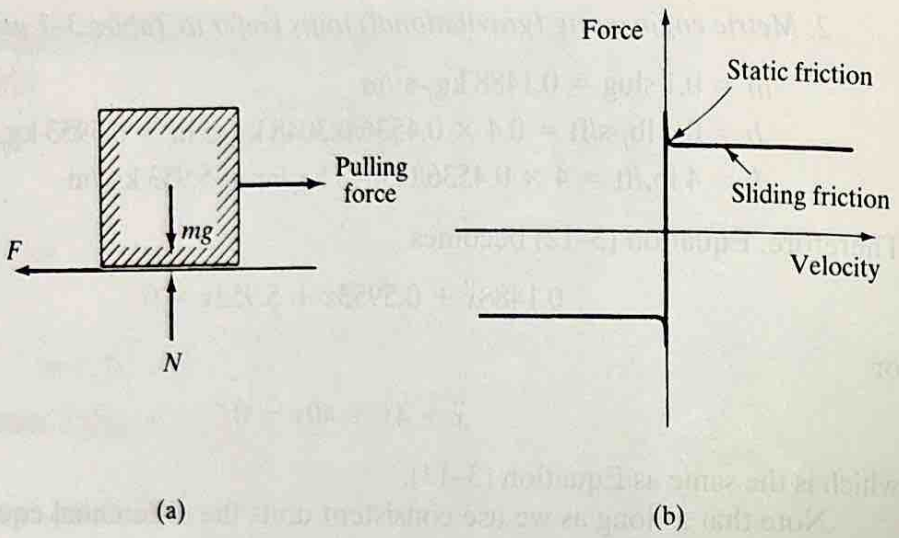


Figure 3–13 (a) Body placed on a rough surface and influenced by a pulling force; (b) characteristic curve for static and sliding friction.

The magnitude N of the normal force and the magnitude F_s of the maximum static friction force are proportional to each other. The ratio F_s/N is called the *coefficient of static friction* and is denoted by μ_s or

$$\mu_s = \frac{F_s}{N}$$

The actual force F of static friction can have any value between zero (when no applied force is parallel to the surface) and a maximum value $\mu_s N$, or

$$0 \leq F \leq \mu_s N$$

If the friction force is that observed for uniform motion of the body, the ratio F_k/N, where F_k is the magnitude of the friction force during uniform motion, is called the *coefficient of sliding friction* or *kinetic friction* and is denoted by μ_k, or

$$\mu_k = \frac{F_k}{N}$$

Thus, when the body is in motion, the force of sliding or kinetic friction is given by

$$F_k = \mu_k N$$

Note that the maximum static friction is larger than sliding friction; that is,

$$\mu_s > \mu_k$$

The coefficients of static and sliding friction depend primarily on the nature of the surfaces in contact.

Comments. The preceding facts on frictional forces give a macroscopic description of friction phenomena. They are based on experimental studies. That is, they are empirical relations but do not represent fundamental laws. In the following list we summarize the facts about static and sliding friction.

1. Friction force always acts opposite to the direction of actual or intended motion. Until motion of the body takes place, the magnitude of the static friction force is equal to the magnitude of the force acting in the direction of motion.

2. The magnitude of sliding friction is proportional to the magnitude of normal force and is nearly independent of the contact area.
3. The coefficient of sliding friction varies somewhat with the relative velocity but may be considered constant over a wide range of speeds.
4. For a given pair of surfaces, the maximum static friction is larger than sliding friction.

Example 3–5

For the system shown in Figure 3–14, let us obtain the force F_i needed at the end of the lever to keep the brake drum from rotating. Assume that the coefficient of the static friction μ_s is 0.4.

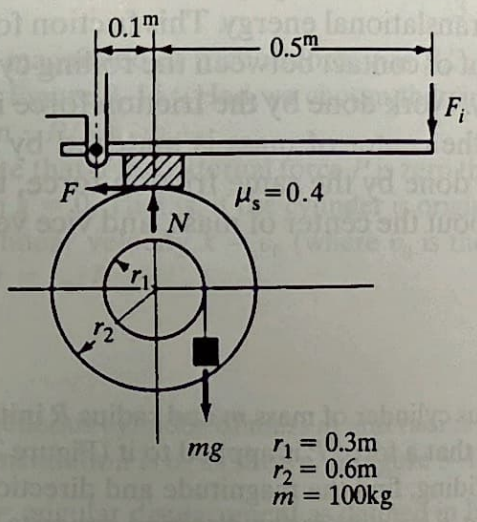


Figure 3–14 Brake system.

The torque due to the weight mg is clockwise, and its magnitude T_1 is

$$T_1 = mgr_1 = 100 \times 9.81 \times 0.3 \frac{\text{kg-m}^2}{\text{s}^2} = 294.3 \text{ N-m}$$

The friction force acting on the brake drum is

$$F = \mu_s N = 0.4 \times 6F_i = 2.4F_i \text{ N}$$

The torque due to the brake force F is counterclockwise and its magnitude T_2 is

$$T_2 = Fr_2 = 2.4F_i \times 0.6 = 1.44F_i \text{ N-m}$$

If $T_2 > T_1$, then the brake drum will not rotate, and so

$$1.44F_i > 294.3$$

or

$$F_i > 204.4 \text{ N}$$

Thus the magnitude of force F_i needed to keep the brake drum from rotating is greater than 204.4 N.

Rolling and sliding motion. Consider the motion of a homogeneous cylinder of mass m and radius R rolling down an inclined plane. Questions of rolling and/or sliding immediately involve questions of friction between the surfaces in